higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 2

(6021032)

22 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 8 pages.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat and in proportion.
5. Start each question on a NEW page.
6. Write neatly and legibly.

Copyright reserved

QUESTION 1

1.1 Choose an item or word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–F) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Precheck before using the micrometer	A	demarcated area
1.1.2	Unit of angular measurement	B	radians
1.1.3	Gas storage area	C	unsafe condition
1.1.4	Smoke from burning oil	D	unsafe action
1.1.5	No safety guard on the machine	E	calibrate
		F	pollution

(5 × 1) (5)

1.2 Name the safety sign that should be displayed in the following motor workshop areas:

1.2.1 At a pedestal drill

1.2.2 At the entrance to the workshop

(2 × 1) (2)

1.3 Give TWO precautionary measures that can be taken to prevent workers from slipping on workshop floors. (2)

1.4 Name THREE parts of the angle grinder in FIGURE 1 below. Write down the letter (A–E), followed by your answer, in the ANSWER BOOK.

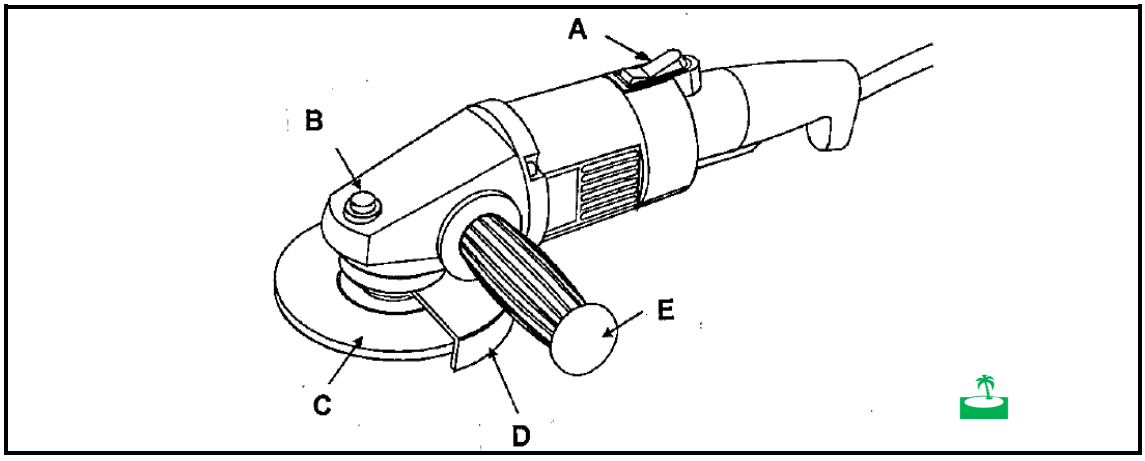


FIGURE 1 (3)

1.5 Paying attention to your own hygiene is very important.

Describe THREE tasks that you can perform to promote personal hygiene in the workshop. (3)

1.6 Name TWO visual inspections that must be done on a bench-grinder before using it. (2)

1.7 Briefly explain the influence that temperature can have on a component when it is being measured. (3)

1.8 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.8.1–1.8.5) in the ANSWER BOOK.

1.8.1 The most appropriate lifting equipment to use when doing wheel alignment on a passenger vehicle is a …

A four-post hoist.
B bottle jack.
C two-post hoist.
D engine hoist.

1.8.2 A/An … is NOT a portable lifting device.

A bottle jack
B two-post hoist
C engine hoist
D trolley jack

1.8.3 A saddle forms part of a/an …

A bottle jack.
B two-post hoist.
C four-post hoist.
D engine hoist.

1.8.4 You should not leave a vehicle on a jack, as it could cause the …

A wheels of the jack to break.
B chocks to be damaged.
C jack to collapse.
D vehicle to roll forward or backward.

1.8.5 After lifting a vehicle with a hoist, you must make sure that …

A the hoist can handle the weight of the vehicle.
B the doors of the vehicle are closed.
C there is no excessive weight in the car.
D the hoist is locked.

(5 × 1) (5)

[25]

QUESTION 2

2.1 Explain why it is important to wear the required PPE when working with batteries. (2)

2.2 Which material is used to manufacture battery casings? (1)

2.3 Are the cells inside a 12-V battery connected in series or in parallel? (1)

2.4 List any THREE functions of an engine-lubricating oil. (3)

2.5 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.5.1–2.5.7) in the ANSWER BOOK.

2.5.1 Sulphation of a battery is caused by alternators that overcharge.

2.5.2 A charging battery gives off fumes that are highly explosive.

2.5.3 Checking the electrolyte level forms part of a service.

2.5.4 Weather and temperatures have no effect on the battery output.

2.5.5 Incorrect jumper starting could cause expensive damage.

2.5.6 A hydrometer is used to check the level of the electrolyte.

2.5.7 When buying a new battery, the amp/hour rating should be taken into consideration.

(7 × 1) (7)

2.6 Which additive is added to engine oil to achieve the following outcomes?

2.6.1 Slowing down the oxidation of metals in the engine

2.6.2 Cleaning and neutralising impurities in the oil

2.6.3 Preventing the oil from becoming too thin at higher temperatures

(3 × 1) (3)

2.7 Name TWO oil pumps that are commonly used in motor vehicle engines. (2)

2.8 Explain the term *hydrodynamic lubrication.* (2)

Copyright reserved

2.9 Give ONE word or term for each of the following descriptions. Write only the word or term next to the question number (2.9.1–2.9.5) in the ANSWER BOOK.

anti-freeze; aluminium; thermostat; two-stroke; cooling system

2.9.1 Opens at a certain temperature to allow coolant to flow through

2.9.2 Many parts working together to keep the engine cool

2.9.3 One of its functions is to prevent rust in the system

2.9.4 This type of engine does not require water as a coolant

2.9.5 A material that reduces the mass of a wheel rim

(5 × 1) (5)
[26]

QUESTION 3

3.1 State THREE possible causes of bearing failure. (3)

3.2 Name TWO parts of a roller-bearing that wear out under normal working conditions. (2)

3.3 List THREE advantages of the more modern aluminium alloy wheel rim over the older steel rim. (3)

3.4 The Department of Transport code for tyre markings holds important information.

List THREE details of the tyres that it represents. (3)

3.5 List THREE tasks that should be performed regularly on tyres in order to extend their life span. (3)

3.6 Briefly explain how the radiator cools down the hot coolant that comes from the engine. (3)

3.7 Refer to FIGURE 2 below to answer the following questions (3.7.1–3.7.3) in the ANSWER BOOK:

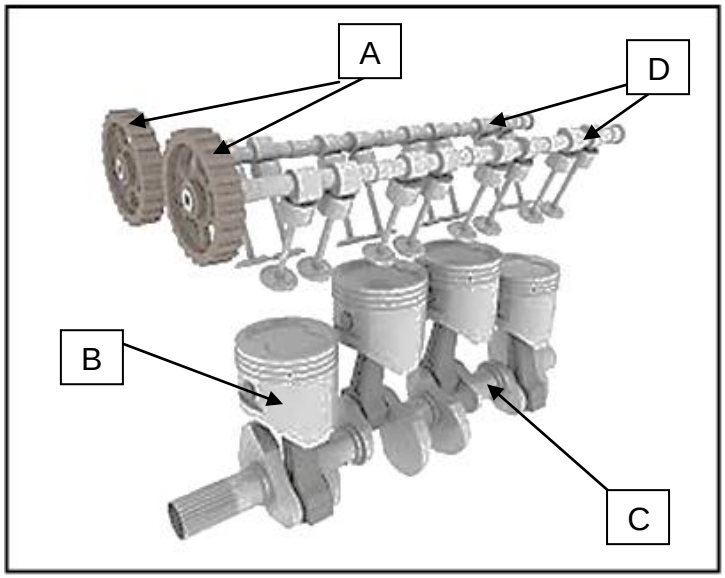


FIGURE 2

3.7.1 Name the parts marked A–D. Write only the answer next to the question number in the ANSWER BOOK. (4)

3.7.2 Name TWO methods of connecting A and C in order to set the valve timing. (2)

3.7.3 State TWO functions of part C. (2)
[25]

QUESTION 4

4.1 FIGURE 3 shows the positions of a headlight beam against a wall. Use the information below to answer the following questions (4.1.1–4.1.2) in the ANSWER BOOK:

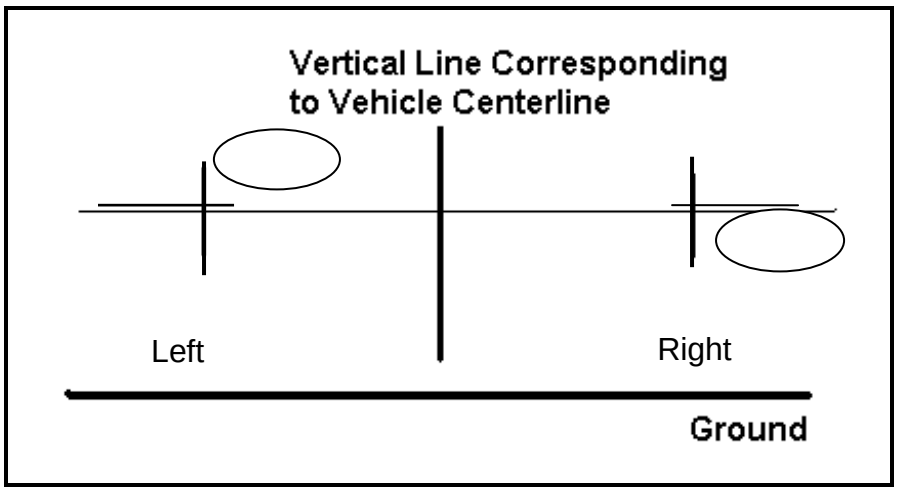


FIGURE 3

4.1.1 What adjustments must be made to the right-side headlight to correct the position on bright beam (high beam)?

4.1.2 What adjustments must be made to the left-side headlight to correct the position on dip/dim beam (low beam)?

(2 × 2) (4)

4.2 State ONE advantage of LED lights over filament-type bulbs. (1)

4.3 List THREE tasks that need to be carried out when preparing a vehicle for a service. (3)

4.4 Name THREE fluid levels other than oil levels that should be checked during a service of a vehicle. (3)

4.5 State THREE reasons why tools should be cleaned and stored after being used. (3)

4.6 Explain why it is important to warm up the engine before you drain the oil. (3)

4.7 The fuel lines are checked under a vehicle on a pre-service inspection.

List the THREE other areas (not items) of the vehicle that should be checked. (3)

4.8 Disposing incorrectly of oil can negatively affect the environment.

Explain how it can cause TWO different types of pollution. (2)

4.9 Name TWO hand tools required to perform a service. (2)

[24]

TOTAL: 100

Copyright reserved